



BIOLOGY CH.22 — MISCELLANEOUS — FULL MIND MAP

Topic-wise · Fully Explained · Hinglish · SSC / BPSC / BSSC / Railway



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1. Plants vs Animals — Structural Differences

- ⚠️ Trap: 'Growth certain regions tak limited nahi hoti Plants mein' — ye statement INCORRECT hai — actually Animals mein growth certain regions (meristematic zones) tak limited hoti hai, Plants mein nahi. Plants move nahi karte, kam energy chahiye hoti hai animals se. [Q1185]

2. Mendel's Success — Methodology & Sample Size

- ⚠️ Trap: Mendel ne 'small sample size' use ki — ye statement INCORRECT hai. Mendel ka success ka main reason tha ek time mein ek hi character study karna, LARGE sample sizes use kiye, aur mathematics se apna work present kiya. [Q1186]
- Mendel's laws sirf tab apply hote hain jab 'parents pure breeding' hon — pea plants consistently same traits wale offspring produce karte the. [Q1285]

- Statement check: Mendel ne pea plant chuna experiments ke liye (TRUE), Chromosomes genetic factors ke carrier hain (TRUE) — lekin human forelimb aur bird wings HOMOLOGOUS organs hain (same structure, different function), Analogous NAHI (FALSE). [Q1309]

3. Mendel's Conclusions — Genes & Gametes

- ⚠ Trap: 'Har gamete mein same gene ki 2 copies hoti hain' — ye statement INCORRECT hai. Har gamete mein har chromosome ki EK copy hoti hai, aur har gene ka ek allele. Genes distinct units ki tarah inherit hote hain, gametes formation ke dauran segregate hote hain. [Q1187]
- ⚠ Trap: 'Law of Biopics' Gregor Mendel ki Law of Inheritance mein SHAAMIL NAHI hai (fictional naam) — 3 asli laws hain: Law of Dominance, Law of Segregation, Law of Independent Assortment. [Q1240]
- Pea plant mein recessive character: 'Wrinkled seed' (aur Green seed, White flower, Terminal flower, Short stem, Constricted pod, Yellow pod) — Dominant: Round seed, Yellow seed, Purple flower. [Q1310]

4. Genetics — A Branch of Biology

- 'Genetics' inheritance patterns ki study karne wali biology branch hai — heredity aur variations ki study karti hai. Gene heredity ki basic unit hai. [Q1188]

5. Reproductive Health & Family Planning

- ⚠ Trap: 'Smoking aur alcohol consumption' family raise karne mein helpful NAHI hai — harmful hai, breast cancer risk badhata hai, fertility affect karta hai, pregnancy mein baby ki health par asar karta hai. [Q1189]
- Teen mothers banne ka possible reason 'Familial aur societal pressure' hai — teenage pregnancy (13-19 age) mein lower educational levels, poverty, family-society pressure related hote hain. [Q1191]
- Young adolescents (10-14 age) mein pregnancy complications ka higher risk hota hai kyunki 'unka body size aksar bahut chhota hota hai baby deliver karne ke liye' — bodies physically fully developed nahi hoti. [Q1207]
- Pregnant woman ke liye best diet 'dietician dwara prescribed special diet' hai — proteins-minerals-vitamins mein rich honi chahiye, calcium-vitamin D-protein important hain. [Q1208]

6. Balanced Diet & Nutrition

- Balanced meal ka example: 'Roti, pulses, vegetables' — carbs (roti), protein (pulses), vitamins-minerals (vegetables) combination hai. [Q1190]

7. Amla — A Medicinal Fruit

- 'Amla' (*Emblica officinalis*) ka fruit Chyavanprash banane mein use hota hai — Unani-Ayurvedic medicine mein important herbal plant hai, vitality-vigor build up karta hai. [Q1192]

8. pH Levels — Body, Acid Rain & Tooth Decay

- pH match: Human body → 7 to 7.8; Acid rain → less than 5.6; Mouth (tooth decay start) → less than 5.5. [Q1193]
- ⚠️ Trap: 'Tooth decay mouth pH 5.5 se HIGHER hone par shuru hota hai' — ye INCORRECT hai — actually LOWER hone par shuru hota hai. Bacteria sugar-food particles degrade karke acids banate hain (base nahi). [Q1204, Q1303]

9. Fermentation — Dough Rising

- Dough (yeast ke saath) ka volume badhta hai kyunki 'CO₂ (respiration ke dauran produced) volume increase karti hai' — gluten proteins CO₂ gas trap karti hain, dough rise karta hai. [Q1194]

10. Gene-Protein Specificity

- Sahi statement: 'Har protein ke liye ek specific gene hota hai' — hormones-enzymes proteins se bane hote hain, ek particular protein ki formation ek particular gene control karta hai (DNA se). [Q1195]

11. Biological Magnification

- Higher trophic levels mein chemicals ka accumulation 'Biological Magnification' kehlaata hai — mercury, arsenic, pesticides (DDT) jaise contaminants — cancer, kidney problems, birth defects cause kar sakte hain. [Q1196]

12. Emulsification of Fats — The Liver

- Fats ke emulsification mein 'Liver' involve hota hai — bile juice complex fat molecules ko chhote molecules mein totti hai, digestion easy banati hai. [Q1197]

13. Nitrogen — Essential for Protein Synthesis

- 'Nitrogen' plants protein synthesize karne ke liye use karte hain — amino acids ka essential constituent hai, soil se nitrate-ammonium form mein roots se absorb hota hai. [Q1198]

14. Blood Plasma — Transport

- 'Plasma' food, carbon dioxide aur nitrogenous wastes ko dissolved form mein transport karta hai — blood ka liquid component hai, hormones-nutrients-metabolic wastes transport karta hai. [Q1199]

15. Respiratory Pigment — Haemoglobin

- ⚠️ Trap: 'Respiratory pigment WHITE blood corpuscles mein present hai' — ye INCORRECT hai — RED blood cells mein hota hai (Haemoglobin). Pyruvate lactic acid (3-carbon) mein convert hoti hai, alveoli walls mein blood-vessels ka network hota hai. [Q1200]
- 'Haemoglobin' pigment ki oxygen ke liye high affinity hoti hai — heme pigment + globin protein se bana hai, RBC mein hota hai. Myoglobin muscles mein oxygen store karta hai. [Q1205]

16. Natural Sources of Acids

- 'Sour milk/Curd' Lactic acid ka natural source hai — Tomato mein Oxalic acid, Ant sting mein Methanoic (Formic) acid, Tamarind mein Tartaric acid, Orange mein Citric acid hoti hai. [Q1201]

17. Phloem — Seasonal Sugar Transport to Buds

- 'Spring' season mein root/stem mein stored sugar buds tak transport hoti hai (jo energy chahiye grow karne ke liye) — Phloem ye transport karta hai, naye buds spring mein bante hain. [Q1202]

18. Why Trees Shed Old Leaves

- Trees purani leaves shed karte hain kyunki 'cells waste materials se pack ho jaate hain' — deciduous trees (Maples, oak) is wajah se leaves girate hain, winter mein cells survive nahi kar paate. [Q1203]

19. Ecosystem — Abiotic-Biotic Interaction

- 'Earthworm ka soil mein burrow banana' abiotic (soil) aur biotic (earthworm) component ki interaction dikhata hai — ecosystem living-non-living components ka interaction hai. [Q1206]

20. Maintenance Processes in Life

- Maintenance processes life mein 'damage aur breakdown prevent' karne ke liye zaroori hain — energy ka source food hai, nutrition food lene ki process hai, growth-repair ke liye raw material bahar se chahiye. [Q1209]

21. Destarching — Dark Room Experiment

- Plants ko dark room mein 3 din rakhne par 'starch resources use ho jaate hain' — photosynthesis rukti hai light na hone se, stored starch respiration-transportation ke liye use hota hai (Destarching kehlaata hai). [Q1210]

22. Xylem — Coloured Water Experiment

- Cut twig ko coloured water mein rakhne se water ka conduction xylem se hota hai (gravity ke against) — twig cut karne par coloured tissue INSIDE (xylem region) mein dikhti hai. [Q1211]

23. Adipose Tissue — Fat Storage

- 'Adipose tissue' fat-storing tissue hai body mein — Areolar tissue loosely arranged fibres wali connective tissue hai, Vascular tissue (xylem-phloem) water-food conduct karti hai plants mein. [Q1212]
- Areolar tissue skin aur muscles ke beech filler tissue ki tarah kaam karti hai — organs ke beech spaces fill karti hai, internal organs support karti hai, tissue repair mein help karti hai. [Q1291]

24. Taxol — An Anti-Cancer Drug from Yew

- 'Taxol' Yew tree (Taxus Brevifolia) se extract hoti hai — Cancer (Breast, lung, Ovarian) ke liye natural source hai USA mein, best plant-based treatments mein se ek. [Q1213]

25. Autotrophs & Holozoic Organisms

- Simple inorganic sources (CO₂, water) se food banane wale organisms 'Autotrophs' kehlaate hain — Plants, lichens, algae examples hain. Saprophytes dead matter se, Heterotrophs doosre organisms se food lete hain. [Q1214]
- Doosre organisms ko khaane wale organisms 'Holotrophic (Holozoic)' organisms kehlaate hain — ingestion-digestion-egestion involve karte hain. Parasitic organisms host par-andar rehte hain, Saprotrophic dead matter se. [Q1301]

26. Food Chain — Trophic Levels

- Food chain ke 3rd trophic level mein 'Secondary consumers' (carnivores jo herbivores khaate hain) aate hain — 1st Producers, 2nd Primary consumers (herbivores), 4th Top/Tertiary consumers. [Q1215]

27. Food Poisoning

- Food poisoning ka main cause 'microorganism se spoilt food khaana' hai (Salmonella, Listeria, E.coli) — symptoms: Nausea, bloody stool, fever, abdominal pain. [Q1216]

28. Genetic Engineering — The Gene Revolution

- 'Gene Revolution' genetic engineering shaamil karti hai — organism ke genes ka direct manipulation biotechnology ke through, food production mein biotechnology ka application hai. [Q1217]

29. Endemic Species

- 'Andaman Teal' endemic species ka example hai — sirf ek geographical region mein exist karti hai. India ke endemic species: Asiatic Lion, Kashmir Stag, Lion-Tailed Macaque, Sangai Deer, Nilgiri Tahr. [Q1218]
- Endemic species wo hain jo 'ek particular area mein exclusively paayi jaati hain' — Galapagos tortoise, Hawaiian honeycreeper, Australian Kangaroo examples hain. [Q1249]

30. Charles Darwin — Galapagos & Theory of Evolution

- 'Charles Darwin' Galapagos Islands se associate hain — fearless wildlife, Darwin's Theory of Evolution ke liye inspiration source. Carl Linnaeus Father of Taxonomy, Ernst Haeckel ne 'Ecology' word coin kiya. [Q1219]
- Charles Darwin ne 'Origin of Species' book (1859) mein organic evolution ka mechanism suggest kiya — 'Father of Evolution' kehlaate hain. Carl Woese 6-kingdom, Whittaker 5-kingdom classification diye. [Q1280]

31. The Human Eye — Parts

- ⚠️ Trap: 'Cochlea' human eye ka part NAHI hai — ye INNER EAR mein hoti hai (hearing ke liye). Eye ke parts: Ciliary muscles, Cornea, Iris, Pupil, Lens, Optic nerve. [Q1220]

32. Cyclone-30 — Cancer Treatment

- 'Cyclone-30' particle accelerator, Cancer patients ke liye help karta hai — VECC (Kolkata) mein operational hai, radioisotopes produce karta hai diagnostic-therapeutic use ke liye. [Q1221]

33. Nobel Prize — Helicobacter Pylori Discovery

- 'Robin Warren aur Barry Marshall' ne Helicobacter Pylori bacterium discover karne ke liye Nobel Prize (Physiology/Medicine) jeeta. [Q1222]

34. GM Crops — Bt Brinjal & Bt Cotton

- Genetically modified brinjal (Bt-brinjal) ka objective tha 'pest resistant banana' — *Bacillus thuringiensis* se Cry1Ac gene insert kiya gaya, Maharashtra Hybrid Seeds Company ne develop ki. [Q1223]
- 'Bt Cotton' India mein legally cultivate karne ki permission mila GM crop hai — *Bacillus thuringiensis* bacterium se resistance gene liya gaya, lepidopterans-dipterans-coleopterans pests ke against resistant hai. [Q1230]

35. Antibiotics — Facts, Myths & Examples

- ⚠️ Trap: 'Antibiotics ke side effects mein smallpox, filaria, diphtheria shaamil hain' — ye INCORRECT hai — ye alag diseases hain (virus, parasite, bacteria se), antibiotics ke side effects nahi. Antibiotics viruses (cold-flu) ke against effective NAHI hain. [Q1224]
- ⚠️ Trap: 'Paracetamol' antibiotic NAHI hai — mild-to-moderate pain, headache, fever ke liye hai. Erythromycin, Tetracycline, Streptomycin antibiotics hain. [Q1272]
- 'Amoxicillin' penicillin antibiotic hai jo bacteria se ladta hai — Tonsillitis, Pneumonia, ear-nose-skin-urinary tract infections treat karta hai. Penicillin Alexander Fleming ne 1928 mein discover ki. [Q1278]

36. Mosquito Life Cycle

- Mosquito lifecycle ke 4 stages: 'Egg, Larva, Pupa, Imago' — female mosquitoes water bodies mein eggs deti hain, larva-pupa transform hokar adult (imago) mosquito banta hai. [Q1225]

37. Nematodes — Missing Body System

- ⚠️ Trap: 'Respiratory system' Nematodes mein NAHI hoti — Reproductive, Excretory, Digestive systems hoti hain, breathing diffusion se hoti hai. Nematodes (roundworms) plants-humans ko infect karte hain. [Q1226]

38. Magnetic Fields — Heart & Brain

- Body mein 'Heart aur Brain' significant magnetic field produce karte hain — MRI brain images ke liye, ECG heart ki electrical activity record karta hai. [Q1227]

39. Vitamins in Yeast

- 'Vitamin B' (especially B-complex, riboflavin B2) yeast mein rich hoti hai — yeast eukaryotic single-celled fungi hai. [Q1228]

40. Penicillium — A Fungus

- 'Penicillium' fungi hai — antibiotic penicillin secrete karta hai (bacteria growth kill-inhibit karta hai). Alexander Fleming ne 1928 mein penicillin discover ki. [Q1229]

41. Quinine — Anti-Malarial Drug from Cinchona

- Anti-malarial drug 'Quinine' Cinchona plant se extract hoti hai — Malaria Plasmodium parasite se hoti hai, Anopheles mosquito ke bite se failti hai. [Q1231]

42. Beneficial & Harmful Insects

- ⚠️ Trap: 'Termite' beneficial insect NAHI hai — destructive pest hai, wood-organic materials khaake buildings damage karta hai. Silk moth, Lac insect, Honey bee beneficial hain. [Q1232]
- 'Wasps aur Beetles' ke teeth (jaws — mandibles-maxillae) hote hain — powerful food-grinding jaws, Phylum Arthropoda ke hain. Bees pollination mein important role play karte hain. [Q1254]

43. Lactose — Intolerance & Composition

- Lactose intolerance wo condition hai jisme 'small intestine dairy products ki sugar (lactose) todne mein unable hoti hai' — Lactose glucose-galactose units se bani disaccharide hai, milk ka 2-8% hoti hai. [Q1233]
- Lactose 'Galactose' aur glucose se compose hoti hai — hydrolysis par ye 2 sugars mein toot jaati hai. [Q1256]

44. The Macula — Area of Best Vision

- Human eye mein 'yellow' oval spot best vision ka area hai — Macula (yellow spot) retina ke center ke pass hota hai, visual acuity ke liye specialized hai. [Q1234]

45. Red Panda — Bamboo Diet

- Red Panda 'Bamboo' leaves khaata hai — roots, succulent grasses, fruits, insects bhi khaata hai. Eucalyptus koala ka main food hai. [Q1235]

46. Normal Human Body Temperature

- Normal human body temperature '98.6°F' (37°C) hai — Clinical thermometer se measure hoti hai, range 35°C-42°C hoti hai. [Q1236]

47. Anthocyanins — Fruit & Vegetable Colours

- 'Anthocyanins' pigments fruits-vegetables ke colours determine karte hain — plant cell vacuoles mein hote hain, water-soluble, pH ke saath color change karte hain. [Q1237]

48. Homologous vs Analogous Organs

- 'Homologous' organs same basic structural design-origin rakhte hain lekin different functions perform karte hain — example: forearm of frog, lizard, bird, human. Analogous organs similar function, different structure-origin (bird-insect wings). [Q1238, Q1293]
- ⚠ Trap: 'Chordates diploblastic hote hain' — ye WRONG statement hai — chordates TRIPLOBLASTIC hote hain (3 germ layers). Chordates coelomate hote hain, dorsal nerve cord aur notochord rakhte hain. [Q1282]
- Analogous organs ka example: 'Tailfin of lobster aur flukes of whale' — functionally similar, structurally different. Human arm-bull's leg, bat's wing-whale's fin, dog's leg-dolphin's flipper — sab HOMOLOGOUS hain. [Q1283]

49. Speciation & Genetic Evolution

- ⚠ Trap: 'Divergent evolution' speciation mein contribute NAHI karta — ye speciation ka RESULT hai, cause nahi. Speciation factors: Genetic drift, Natural selection, Geographic isolation, Variation, Mutation. [Q1239]
- Species ke andar 'change' (variations) survival advantage de sakti hai ya genetic evolution mein contribute kar sakti hai — natural selection se adapted organisms survive-genes pass karte hain. [Q1284]

50. Genome

- Organism ke complete nuclear DNA set ko 'Genome' kehte hain — Tissue cells ka group hai, Organelles cell ke andar structures hain, Chromosomes DNA-protein se bane structures hain. [Q1241]

51. Hydroponics

- 'Hydroponics' farming mein plants mineral nutrient solutions mein bina soil ke grow karte hain — Tomatoes, Lettuce, Cucumber, Banana, Spinach jaise plants isse grow hote hain. [Q1242]

52. Omnivores — Plants & Animals Dono Khaane Wale

- Plants AUR animals dono khaane wale animals 'Omnivores' kehlaate hain — Bears, Birds, Dogs, humans examples. Carnivores meat, Herbivores plants khaate hain. *[Q1243]*

53. Diffusion

- 'Diffusion' se molecules high concentration se low concentration ki taraf move karte hain — Example: Oxygen plasma membrane ke across move karta hai. *[Q1244]*

54. Dental Caries — Tooth Decay

- 'Dental caries' ka matlab 'Tooth decay' hai — mouth mein rehne wale bacteria acids produce karke teeth ko attack karte hain. *[Q1245]*

55. Alfalfa — Sativa

- 'Sativa' (*Medicago sativa*) ko 'Alfalfa' kehte hain — legume crop hai, forage-hay-cover crop ke liye grow ki jaati hai. *[Q1246]*

56. Nipah Virus — Fruit Bats

- 'Fruit bats' (flying foxes) Nipah virus ke natural hosts hain — pigs-humans ko bhi illness cause kar sakta hai, encephalitis associate hai. *[Q1247]*

57. Human Evolution — Genetic Roots in Africa

- Human species ki genetic roots 'Africa' mein hain — *Homo naledi* fossils South Africa mein mile. Evolution stages: *Dryopithecus* → *Ramapithecus* → *Australopithecus* → *Homo Erectus* → *Homo Sapiens*. *[Q1248]*

58. Wool — Shearing Sheep

- Sheep ki fleece thin skin layer ke saath remove karne ki process 'Shearing' kehlaati hai — wool obtain karne ka process: Scouring, Sorting, Removing burrs, Cleaning-drying. *[Q1250]*

59. Falcon — Speed

- Falcon (*Peregrine Falcon*) ki approximate speed '320 km/h' hai — prey 2 miles (3.2 km) door se spot kar sakta hai. *[Q1251]*

60. Groundnut — An Oil-Yielding Plant

- 'Groundnut' oil-yielding plant hai (Arachis hypogaea) — 'kingpin of oilseeds' kehlaata hai, 85% India mein kharif season mein sown hoti hai. [Q1252]

61. Iodine Test — Detects Starch

- 'Iodine' food mein 'Starch' ki presence test karne ke liye use hoti hai — starch presence mein iodine solution blue-black color mein badal jaati hai. [Q1253]

62. Jasmine — A Creeper Plant

- 'Jasmine' (Jasminum) creeper plant hai — small vine-like plants ground ke close grow karte hain. Rose 'King of flowers' kehlaata hai. [Q1255]

63. Dendrimers — Pharmaceutical Nanotechnology

- 'Dendrimers' nano-sized, radially symmetric molecules Health-Pharmaceutical industry mein use hote hain — anti-cancer drugs, transdermal drug delivery mein applications. [Q1257]

64. Phototropism

- 'Phototropism' ka matlab hai 'plants ka light ki taraf movement' — shoot growth ko light source ki direction mein reorient karne ki ability. Positive-Negative phototropism hoti hai. [Q1258]

65. Rhizome, Tuber, Bulb & Rhizoid — Underground Structures

- 'Rhizome' segmented stem hai jo horizontally underground grow karta hai (Ginger, Poison Oak, Hops) — Tuber underground grows (Potato), Bulb short stem with fleshy leaves (onion, garlic). [Q1259]
- ⚠️ Trap: Rhizoids 'multi-cellular aur epithelial tissues se composed' hote hain — ye INCORRECT hai. Mosses mein multicellular, liverworts-hornworts mein UNICELLULAR hote hain, epithelial tissue se NAHI bane. Root hairs jaisa function karte hain bryophytes mein. [Q1296]

66. Prunes — Dried Plums

- 'Prunes' European plum (Prunus domestica) se obtain hote hain — dried plums hain, digestion help karte hain, potassium-vitamins-iron rich hote hain. [Q1260]

67. Eutrophication

- Lakes mein excessive Nitrogen-Phosphorus ki condition 'Eutrophic' kehlaati hai — plants-algae ki growth badhti hai, dissolved oxygen kam hota hai (harmful aquatic life ke liye). [Q1261]
- Algal bloom se 'Eutrophication' hoti hai — excessive minerals-nutrients ki wajah se algae excessive grow karti hai. [Q1270]

68. Banana — Edible Endocarp & Mesocarp

- Banana ka edible part 'Endocarp aur Mesocarp' hai — Epicarp outer skin hai, Mesocarp fleshy middle layer, Endocarp innermost layer (seeds contain karta hai). [Q1262]

69. Ectoparasites vs Endoparasites

- 'Mosquito' ectoparasite ka example hai — host ke BAHAR rehta hai (flea bhi). Endoparasites (tapeworm, hookworm) host ke tissues-organs ke ANDAR rehte hain. [Q1263]

70. Blood Sugar — Glucose

- 'Glucose' ko 'blood sugar' kehte hain — 6-carbon compound hai, hydrogen-oxygen contain karta hai. [Q1264]

71. Lipids — Glycerol & Fatty Acids

- Lipids tab bante hain jab glycerol 'Fatty acids' se combine hoti hai — Oligosaccharides carbohydrate chains hain, Peptide amino acids ki short chain hai. [Q1265]

72. Pulse Beat — Measured from Artery

- Pulse beat 'Artery' se measure hoti hai — circulatory system ka part hai, oxygen-rich blood heart se body cells tak le jaati hai. [Q1266]

73. Brahmi — Ayurvedic Memory-Boosting Herb

- 'Brahmi' (Bacopa monnieri) Ayurveda mein memory improve karne ke liye use hoti hai — bacosides compounds cognitive function-memory-learning improve karte hain. [Q1267]

74. Sarcolemma — Muscle Fiber Membrane

- 'Sarcolemma' muscle fiber (single muscle cell) ko cover karne wali membrane hai — physical forces control karne mein help karti hai. [Q1268]

75. Transpiration

- Forests atmosphere ki humidity 'Transpiration' se badhate hain — plants soil se water lekar leaves se water vapour release karte hain. [Q1269]
- Plants mein water vapour ki form mein water loss ko 'Transpiration' kehte hain — roots se water absorb karke stomata se release hoti hai, plant ko cool bhi karti hai. [Q1295]

76. Plasmolysis

- Living plant cell ka water lose karke shrink hona (osmosis se) 'Plasmolysis' kehlaata hai — Cytolysis cell burst hona hai (excess water se), Hemolysis RBC breakdown hai. [Q1271]

77. Vocal Cords — Human Sound Production

- Human sounds 'Vocal cord' mein vibrations se produce hote hain — larynx (voice box) mein hote hain, lungs se air pass hone par vibrate karte hain. [Q1273]

78. Edward Jenner — Father of Immunology

- 'Edward Jenner' smallpox vaccine develop ki, 'Father of Immunology' kehlaate hain — Antoni van Leeuwenhoek Father of Microbiology, Robert Koch ne TB-cholera ke organisms discover kiye. [Q1274]

79. First-Aid Medicines — Ibuprofen & Antihistamines

- 'Ibuprofen' pain, fever aur inflammation treat karne ke liye liya jaata hai — NSAID (Non-steroidal Anti-inflammatory Drug) hai. [Q1275]
- 'Antihistamine' medications hay fever aur doosri allergies ke symptoms kam karne ke liye li jaati hain — Antacids indigestion-heartburn ke liye, Bronchodilators asthma ke liye hote hain. [Q1276]

80. Carcinogens

- 'Carcinogen' cancer directly cause karne wala substance hai — chemical, virus ya radiation ho sakta hai. Common carcinogens: Tobacco, Radon, Asbestos, UV Rays, Alcohol. [Q1277]

81. Heredity — Trait Transmission

- Parents se offspring tak physical-mental traits ka transmission 'Heredity' kehlaata hai — progeny parents jaisi dikhti hai lekin identical nahi hoti (variation ki wajah se). [Q1279]

82. Variations — Inherited During Reproduction

- Reproduction ke dauran arise hone wali variations 'Inherited' ho sakti hain — genetic recombination, mutations, aur doosre factors ka result hoti hain. [Q1281]

83. Somatic Mutation — Not Hereditary

- 'Somatic' mutation hereditary NAHI hoti — body ke cells mein hoti hai (reproductive cells chhodkar), isliye offspring tak transmit nahi hoti. Deletion, Insertion, Duplication doosre mutation types hain. [Q1286]

84. Regeneration — Not the Same as Reproduction

- 'Regeneration' reproduction ke same NAHI hai — Budding, Fission, Multiple fission reproduction ke modes hain, Regeneration repair-regrowth hai (Hydra, Planaria examples). [Q1287]
- Regeneration ka matlab hai: 'lost part ka replacement' AUR 'small parts se poore body ki formation' — dono options sahi hain (Hydra-Planaria examples). [Q1299]

85. Vascular System — Fluid Transport

- 'Vascular' system ek jagah se doosri jagah fluids transport karta hai — plants mein water-nutrients, animals mein cardiovascular system ki tarah, resources deliver karta hai aur mechanical support deta hai. [Q1288]

86. Species & Kingdom — Classification Extremes

- 'Species' living organisms ki classification ki basic unit hai — similar characteristics wale individuals ka group, interbreed karke fertile offspring bana sakte hain. [Q1289]
- Classification hierarchy mein 'Kingdom' grouping mein sabse zyada organisms hote hain common characteristics ke saath — highest category hai biological classification mein. [Q1297]

87. Cloning

- 'Cloning' se cell/organism ki exact copy production hoti hai — Dolly the sheep pehla cloned mammal thi (Ian Wilmut, Roslin Institute). Fertilization male-female sex cells ka fusion hai. [Q1290]

88. Evolutionary Progress — Gradual Change

- Primitive organisms mein millions of years mein gradual changes ki sequence, jo naye species banati hai, 'Evolutionary progress' kehlaati hai — Analogous/Homologous organs alag concepts hain. [Q1292]

89. Tooth Enamel — Calcium Phosphate

- Tooth enamel 'Calcium phosphate' se bana hota hai — hydroxyapatite crystal structure mein hota hai, body ka hardest substance hai, pH 5.5 se neeche corrode hota hai. [Q1294]

90. Muscular Force & Joints

- Muscles ki action se generate hone wali force 'Muscular force' kehlaati hai — joints (fulcrum ki tarah) ke through movement karte hain. Joints 3 types: fibrous, cartilaginous, synovial. [Q1298]

91. Smooth Muscle Tissue — Locations

- Smooth (unstriated) muscle tissue 'Digestive system aur Artery' mein hoti hai — hollow visceral organs ki walls mein, autonomic nervous system se control hoti hai, slow-involuntary contraction. [Q1300]

92. Contractile Proteins — Actin & Myosin

- Muscles mein special 'Contractile proteins' hote hain — thick-thin myofilaments (myosin-actin) se bane hote hain, muscle contraction sliding filament theory se explain hoti hai. [Q1302]
- 'Contractile protein' (Actin aur Myosin) muscle tissue ke contraction-sagging mein help karti hai jisse movement hota hai — Actin thin filaments, Myosin thick filaments. [Q1304]

93. Glycogen — Energy Storage

- Food se derived energy ka kuch hissa 'Glycogen' ki form mein store hota hai — muscles-liver mein store hokar zaroorat par energy mein convert hota hai. [Q1305]

94. Genotype-Phenotype — The ttRr Example

- Genotype 'ttRr' ka phenotype 'Dwarf aur Round' hoga — 'tt' Dwarf (recessive homozygous), 'Rr' Round seeds (dominant allele R). [Q1306]

95. Tendon — Collagen & White Fibres

- 'Tendon' mein abundant white fibres (collagen) hoti hain — muscles ko bone se connect karti hai. Types: Flexor (bend joints), Extensor (straighten joints). [Q1307]
- Tendon 'sirf collagen fibres' se bana hota hai — muscles ko bone se connect karta hai (fibrous connective tissue). [Q1311]

96. Reproduction — Maintaining Species

- 'Reproduction' species ke individuals ki number maintain karne aur extinction rokne ke liye zaroori hai — biologically similar offspring produce karti hai. [Q1308]